

A student team building an autonomous boat from the ground up.

One composite planing hull, two motor pods steering by differential thrust, and a rudder for stability at speed. Designed as one system and built by students at UW–Madison.



WHAT MAKES US DIFFERENT

Designed as one system

One optimization model picks hull shape, drive, battery and speed together, so no part is sized alone.

Industry experience

Members have worked at Tesla, Northrop Grumman, Xcel Energy, Milwaukee Tool, John Deere and Fincantieri.

Built in-house

The composite hull, high-voltage power system, firmware and autonomy all come from one student team.

OUR APPROACH

01

Hull

Plane well, carry the load, stay light.

- 4–6 ft rigid composite hybrid planing hull
- Shape tuned by the optimization model
- Resistance checked in CFD first

02

Electric

Spend every watt-hour on speed.

- Battery storage built with high-voltage safety practice
- Motor pods and propellers sized by the model
- Differential thrust steers, rudder adds stability

03

Autonomous

Find the way and steer, with no one at the helm.

- GPS, IMU and LIDAR sense position and obstacles
- Firmware runs the motors and protects the battery
- Controls software plans the route and steers

TIMELINE

Oct 25 Hull design	Nov 15 Electrical design	Nov 25 Hull built	Jan 25 First drive	Feb 15 First fast drive	Mar 15 Tuning and testing
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